

# Mark Jarzynski

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## SUMMARY

- **Experienced Graphics Engineer** - Experienced graphics engineer with over 15 years of software development experience with a focus on graphics, content pipeline, and optimization.
- **Published Graphics Research** - Published graphics researcher with three papers in major publications, including a highly referenced paper on hash functions quality/speed trade-offs and a new class of performant hash functions tuned for multidimensional values.
- **Director & Manager** - Managed a wide range of team sizes and skills over 20+ research projects, including graphics optimization and a geospatial visualizer for Unreal Engine 4.
- **Excellent Cross-Discipline Communicator** - Demonstrated ability to maintain positive and productive communication with other disciplines, including art and design.

## SKILLS

**Languages:** C • C++ • HLSL • GLSL • C# • Python • Java • Php • HTML • CSS • JavaScript

**Graphics APIs:** Direct X 11/12 • Vulkan • OpenGL

**Graphics Debuggers:** PIX • Nvidia Nsight Graphics/Aftermath • AMD Radeon GPU Profiler

**Game Engines:** Unreal Engine • Unity • Nitrous • Iron Engine

**Platforms:** Windows • Linux • macOS • Android • iOS • Oculus • SteamVR

**Databases:** MariaDB • MySQL • MSSQL • Postgres • Oracle

**Source Control:** Perforce • Git • Subversion • CVS • Mercurial

**Tools:** Visual Studio • VS Code • XCode • CMake • Makefiles • GCC • LLVM • ImGui • OptiTak • NGINX • Apache • ImGui • Docker • JIRA • Confluence • Azure DevOps • Google Workspace

## Experience

### Graphics Engineer - Oxide Games

2021 - present

- Extended and maintained the Decoupled Shading Engine of Nitrous, Oxide's custom game engine used for Ara: History Untold and Ashes of the Singularity II.
- Developed and maintained Nitrous' graphics features and shader, including terrain, water, point lights, area lights, effects, particles, trails, and a hybrid BRDF model (GGX + OrenNayar).
- Worked closely with design and art departments to achieve the desired VFX, lighting, and game visuals and to ensure tools were easily usable via ImGui inferences and included debug features.
- Optimized Nitrous' terrain shader using an expanded stochastic hex blending tiling approach, resulting in a highly flexible terrain tool with no apparent repeats and over 50% increased GPU throughput, enabling it to operate on limited systems including the Steam Deck.
- Created dynamic world terrain modifications to create flat building foundations and smooth transitions to city infrastructure. Implemented De Casteljau's algorithm to provide artistic control on Terrain slopes globally, while preserving mountain peaks.

- Reimplemented ton mapping system with new techniques that better preserve hue in HDR colors and a supplementary interpolation between new and old techniques for artists to select desired visuals.
- Developed a dynamic infrastructure renderer through extensive collaboration with art and design to enable real-time city generation with live debugging, cinematics support, advanced building and terrain aware placement, with destruction and deterioration modifications.
- Wrote SSE-optimized OBB collision to the graphics acceleration structure to efficiently enable long light beam visuals.
- Created deterministic pseudo random number generators that are now used studio-wide in all projects for repeatable generation.
- Added status condition system to efficiently handle status state and logic across a large number of units.
- Implemented an anisotropic BRDF to support shading hair, metal, and fur.

#### **Technical Director - UMBC, Imaging Research Center**

2014 - 2021

- Led, supervised, and educated teams of up to 13 students and staff on a variety of projects.
- Led 20+ research projects with focuses including ray tracing, hash functions, parallel computing, procedural generation, motion capture, photogrammetry, data visualization, and XR.
- Maintained and supported a 10x10 meter VR space and 100 camera photogrammetry facility to support various research projects.
- Equipped teams with the best tools and resources available to support their projects, personal development, and future career opportunities.

#### **Systems Administrator & Programmer - UMBC, Imaging Research Center**

2011 - 2014

- Developed and maintained mission critical software to meet research needs.
- Ensured researchers' needs were being met by student teams through leadership and mentorship.
- Maintained department's infrastructure, servers, and desktops.

## Published Games

[Ashes of the Singularity II](#)

TBA

[Sins of a Solar Empire II: Paths to Power](#)

2025

[Ara: History Untold](#)

2024

## Publications

[Generalized Decoupled and Object Space Shading Systems](#), EGSR

2022

[Hash Functions for GPU Rendering](#), JCGT

2020

[Extending CoNavigator into a Collaborative Digital Space](#), GROUP '20

2020

## Education

**Ph.D in Computer Science**; University of Maryland, Baltimore County

2024 - present

Dissertation: *Hashing Out Performance, Evaluation and Optimizing Hash Functions for GPU Rendering*

**B.S. in Computer Science**; University of Maryland, Baltimore County

2011